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FEDERAL SPECIFICATION

REAMERS, HAND AND MACHINE: ARBORS, SHELL REAMER: AND PILOTS, ADJUSTABLE HAND REAMER

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand reamers, machine reamers, arbors for shell reamers, and pilots for hand adjustable reamers.

1.2 Classification.

1.2.1 Types, classes, and styles. Reamers, arbors, and pilots covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I - Hand.

- Class 1 - Straight fluted.
- Class 2 - Helical fluted.
- Class 3 - Engine valve guide, solid.

Type II - Expansion (hand).

- Class 1 - Straight fluted.
- Class 2 - Helical fluted.

Type III - Adjustable, inserted blade.

- Class 1 - Straight fluted (hand).
- Class 2 - Helical fluted (hand).
- Class 3 - Self-aligning (hand).
- Class 4 - Chucking, straight shank (machine).
- Class 5 - Chucking, taper shank (machine).

Type IV - Pilots (for type III, class 1 adjustable reamers).

Type V - American National Standard taper socket (Morse).

- Class 1 - Straight shank (hand).
  - Style A - Finishing.
  - Style B - Roughing.

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Class 2 - Taper shank (machine).

Type VI - Taper, pipe.

Type VII - Buring, pipe.

Class 1 - Ratchet type (hand).
Class 3 - Bit brace shank (hand).
Class 4 - Straight shank (machine).

Type VIII - Taper pin.

Class 1 - Straight fluted (hand).
Class 2 - Spiral fluted (hand).
Class 3 - High spiral fluted (machine).

Type IX - Bridge, helical fluted (machine).

Type X - Fluted, chucking (machine).

Class 1 - Taper shank.

Style A - Straight shank.
Style B - Helical fluted.

Class 2 - Straight shank.

Style A - Straight fluted.
Style B - Helical fluted.

Type XII - Jobbers' (machine).

Type XIII - Repairman's T-handle (hand).

Type XIV - Brown and Sharpe taper socket (hand).

Class 1 - Straight fluted.
Class 2 - Spiral fluted.

Type XV - Shell (machine).

Class 1 - Straight fluted.
Class 2 - Left hand, helical fluted.

Type XVI - Arbors (for type XV reamers).

Class 1 - Taper shank.
Class 2 - Straight shank.

Type XVIII - Car reamer, left hand, helical fluted, taper shank (machine).

1.2.2 Grades. Reamers shall be made of one of the following grades of steel, as specified herein. Arbors, pilots, and reamer bodies used to hold blades may be made of alloy steels.

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Grade A - High speed steel.
Grade B - Carbon tool steel.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

PPP-P-40 - Packaging and Packing of Hand Tools

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-66 - Steel: Chemical Composition and Hardenability
FED-STD-123 - Marking for Shipment (Civil Agencies)
FED-STD-358 - Sampling Procedures

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute (ANSI)

ANSI B94.2-83 - Reamers

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more reamers, arbors, and pilots for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Illustrations. The illustrations (figures) shown herein are descriptive and not restrictive. They are included for convenience of identification and are not intended to preclude the purchase of reamers, arbors, and pilots which are otherwise in accordance with the requirements of this specification.

3.3 Material. The reamers, arbors, and pilots shall be of the grade steel specified (see 1.2.2) for the applicable type, class and style (see 1.2.1). The chemical composition of the carbon tool and high-speed steels, together with the heat treatment employed, shall be such as to produce reamers having long use life and the cutting qualities necessary for the class of service for which they are normally used.

3.4 Cutting edges. Except as otherwise specified herein, reamers shall be right hand and ground for reaming steel. The cutting edges of reamers shall be properly relieved and the flutes shall provide chip clearance. Flute spacing shall be in accordance with the manufacturer's standard practice.

3.5 Sizes, dimensions and tolerances. Sizes, dimensions, and tolerances of all reamers and arbors covered by this specification except type I, class 3 reamers; type III, class 1 through 5 reamers; type IV pilots (for type III, class 1 reamers); type VII, classes 1 through 4 reamers; and type XIII reamers shall conform to ANSI B94.2. The shanks of reamers shall be concentric with cutting edges within 0.001 inch total indicator reading (tir) as measured between centers. Reamers shall be ground for reaming steel in accordance with the manufacturer's standard practice. The back taper (larger at front), if any, shall

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be the entire length of flutes with a minus tolerance of 0.0001 to 0.0006 inch. Tolerance on taper rate of taper reamers shall be ±0.002 inch per foot. Taper shanks, pilots, and pilot bushings of reamers shall have a surface roughness of not more than 25 microinches, arithmetical average (aa).

3.6 Hardness. The full length of the cutting edges of all reamers except type I, class 3 (see 3.11.3) and type VII, class 1, grade B (see 3.17.1.5) shall be hardened and tempered to a Rockwell hardness value of not less than C63 and not more than C67. Pilot bushings for type III, class 3 reamers shall be hardened and tempered to a Rockwell hardness value from C45 to C60. Pilot bushings for type III, class 1 reamers (type IV pilots) are soft. Type XVI arbors shall have the following Rockwell C hardness values:

(a) If made of steels hardened throughout:
    (1) Taper driving section (arbors) - 40 to 60.
    (2) Driving lugs - soft to 50.
(b) If made of case hardened steels:
    (1) Taper shank (arbors) - 35 to 65.
    (2) Driving lugs - soft to 65.
    (3) Reamer end - 50 to 65.

3.7 Threads. All reamers provided with expansion adjustment shall have fine threads in accordance with FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.8 Miscellaneous requirements. The tangs of taper shank reamers shall be heat treated to a minimum Rockwell hardness of C30 or be relieved in a manner to reduce to a minimum the possibility of the end being upset. Shanks may be integral or joined to the reamer by welding. Mechanical union, such as threaded construction, shall not be acceptable. The flutes of helical fluted reamers shall be not less than six degrees and not more than 20 degrees with the axis, except type VIII, class 3, reamers shall have a spiral angle of not less than 30 degrees and not more than 50 degrees.

3.9 Performance. Reamers shall cut freely and produce smooth and accurate holes when subjected to the tests specified herein.

3.10 Identification marking. Reamers and arbors shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacturer may be readily determined. Reamers, except for type III adjustable, shall be marked in a plain and permanent manner with their nominal size. Type III, class 1, reamers (blades) shall be marked in sets. Type III, classes 2, 3, and 4, reamers (the blade segments) shall be marked in sets with nominal size. The high-speed steel reamers shall, in addition to the foregoing, be marked in a plain and permanent manner with the words "High Speed" or the abbreviation "HS". The marking shall be done in such a manner that it shall in no way interfere with the proper fit or satisfactory operation of the reamers or arbors.

3.11 Type I, hand. Type I reamers shall have straight round shanks with squared ends. The cutting edges at the point of entrance to the hole shall be slightly tapered for a distance of not more than one-fourth of the length of the flutes.

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3.11.1 Class 1, straight fluted. The reamers shall be grade A steel, have straight flutes, and shall conform to table 3 and the applicable tolerances of ANSI B94.2 for the dimensions specified (see 6.2).

3.11.2 Class 2, helical fluted. The reamers shall be grade A steel, have helical flutes, and shall conform to table 4 and the applicable tolerances of ANSI B94.2 for the dimensions specified (see 6.2).

3.11.3 Class 3, engine valve guide, solid. The reamers shall have straight round shanks with squared ends and shall be of grade B steel. The full length of the cutting edges shall be hardened and tempered to a Rockwell hardness value of not less than C60 and not more than C63. The reamers shall be constructed with a pilot 1/2 ±1/16 inch in length for removing scale and carbon. The cutting section of the pilot shall be slightly smaller than the cutting section of the reamer. Class 3 reamers shall have helical flutes, shall be similar to figure 1, and shall conform to table I.

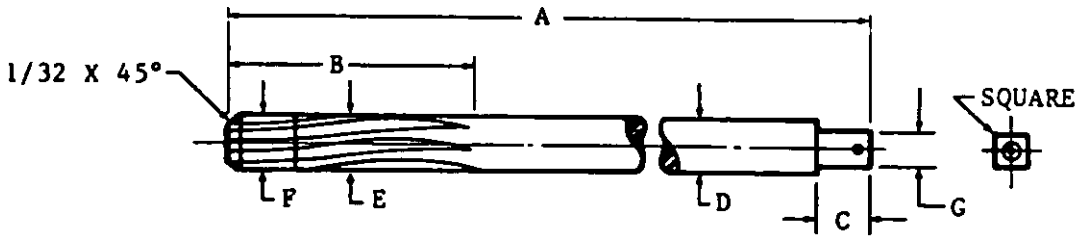


FIGURE 1. Type I, class 3, engine valve guide, solid, helical fluted reamer.

TABLE I. Type I, class 3, engine valve guide, solid, helical fluted reamer.

Diameter (E) +0.001 -0.000	Length overall (A) ±1/16	Length of flutes (B) ±1/16	Length of squares (C) ±1/32	Diameter of shank (D) +0.000 -0.005	Size of Square (G) +0.000 -0.005	Number of flutes	Diameter of pilot (F) +0.000 -0.000
Inch	Inches	Inches	Inch	Inch	Inch		Inch
5/16	7 1/2	1 3/4	5/16	0.307	0.235	6	0.310
21/64	7 1/2	1 3/4	5/16	.323	.250	6	.312
11/32	7 1/2	1 3/4	5/16	.338	.250	6	.342
23/64	8	1 3/4	5/16	.354	.265	6	.343
3/8	8	1 3/4	3/8	.370	.296	6	.373
25/64	8	1 3/4	3/8	.385	.296	6	.375
13/32	8	1 3/4	3/8	.401	.296	6	.404
27/64	8	1 3/4	3/8	.417	.312	6	.406
7/16	9	2 1/4	3/8	.432	.328	8	.435
29/64	9	2 1/4	3/8	.448	.328	8	.437
15/32	9	2 1/4	3/8	.463	.343	8	.467
31/64	9	2 1/4	7/16	.479	.359	8	.469
1/2	9	2 1/4	7/16	.495	.359	8	.498

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3.12 Type II, expansion (hand). Type II reamers shall have straight, round shanks with squared ends. Means shall be provided so that, by the movement of a screw at the bottom end of the reamer, the cutting edges shall be uniformly expanded or contracted within the range for which the reamer is designed.

3.12.1 Class 1, straight fluted. The reamers shall be of grade B steel, have straight flutes, and shall conform to table I and the applicable tolerances of ANSI B94.2 for the dimensions specified (see 6.2).

3.12.2 Class 2, helical fluted. The reamers shall be grade B steel, have left-hand helical flutes, and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.13 Type III, adjustable, inserted blade.

3.13.1 Class 1, straight fluted (hand). The reamers shall be fitted with removable cutting blades made of grade A steel, which shall be adjustable for reaming holes of any size within the range for which the reamer is designated. Reamers shall have straight, round shanks with square ends. The body portion shall contain slots for holding the blades. The reamers shall be constructed to permit an adjustment of not less than 0.035 inch between the minimum and maximum size attainable with any one set of blades. The reamer assembly shall be free cutting and capable of reaming round, straight, and smooth holes free of chatter marks within any size range. When specified, class 1 reamers shall be furnished with pilots (see 3.14 and 6.2). The reamers shall be similar to figure 2 and conform to table II.

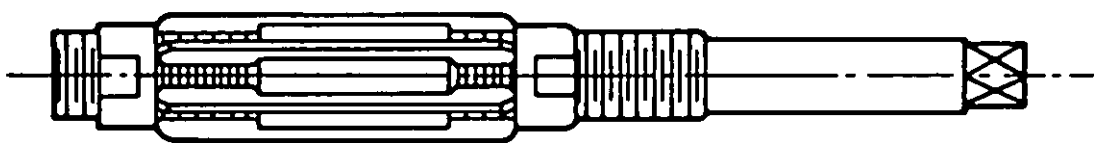


FIGURE 2. Type III, class 1, adjustable, inserted blade reamer, straight fluted (hand).

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TABLE II. Type III, class 1, adjustable, inserted blade reamer, straight fluted (hand)

Nominal size	Range of expansion		Length of blades 1/		Length overall		Number of blades	Thread sizes
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	
Size letters	Inches	Inches	Inches	Inches	Inches	Inches		
8/A	1/4	9/32	1 11/32	1 5/8	3 1/4	3 1/2	4	Optional
7/A	9/32	5/16	1 11/32	1 5/8	3 1/2	3 3/4	4	Optional
6/A	5/16	11/32	1 1/2	1 3/4	4	4 1/8	4	Optional
5/A	11/32	3/8	1 1/2	1 3/4	4 1/4	4 3/8	4	Optional
4/A	3/8	13/32	1 1/2	1 7/8	4 1/2	4 3/4	6	Optional
3/A	13/32	7/16	1 1/2	1 7/8	4 3/4	5	6	Optional
2/A	7/16	15/32	1 5/8	1 7/8	5	5 1/4	6	Optional
A	15/32	17/32	1 1/2	2	4	5 1/2	6	5/16-24
B	17/32	19/32	1 5/8	2 1/8	4 1/2	6	6	3/8-24
C	19/32	21/32	2	2 1/2	5	6 1/2	6	7/16-20
D	21/32	23/32	2	2 3/4	5 1/2	7	6	1/2-20
E	23/32	25/32	2 1/4	3	6	7 3/4	6	9/16-18
F	25/32	27/32	2 1/4	3 1/2	6 1/2	8 1/2	6	5/8-18
G	27/32	15/16	2 3/4	3 1/2	7	9	6	11/16-16
H	15/16	1 1/16	2 3/4	3 1/2	7 1/4	9 3/4	6	3/4-16
I	1 1/16	1 3/16	3	3 7/8	7 3/4	10 3/4	6	7/8-14
J	1 3/16	1 11/32	3 1/8	4	8	11 3/4	6	7/8-14
K 2/	1 11/32	1 17/32	3 3/8		9	12 3/4	6	1-14
	1 11/32	1 1/2	3 3/8		9	12 3/4	6	1-14
L 2/	1 17/32	1 25/32	3 1/2		13	14	6	Optional
	1 1/2	1 15/16	3 1/2		13	14	6	Optional
M 2/	1 25/32	2 3/32	4	5	15	16	6	Optional
	1 13/16	2 7/32	4	5	15	16	6	Optional
N 2/	2 3/32	2 15/32	4 13/16	5 3/4	17	18	6	Optional
	2 7/32	2 3/4	4 13/16	5 3/4	17	18	6	Optional
O 2/	2 15/32	2 31/32	5 3/16	6 1/2	19	20	6	Optional
	2 3/4	3 11/32	5 3/16	6 1/2	19	20	6	Optional

1/ Length of blades to be measured between adjusting nuts with the reamer set at maximum size.

2/ Either range is acceptable. However, when provided in sets, the ranges furnished shall be such that complete dimensional coverage is provided.

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3.13.1.1 Blades. The blades shall be accurately machined and ground in sets to properly fit slots provided on the body of the reamer. Tolerances for the length of individual sets of blades shall not exceed plus of minus 0.0025 inch. The width of the blades shall be uniform for the same size and make of reamer to insure interchangeability. Cutting surfaces of blades shall be concentric with the axis of the body in all positions throughout the range of adjustment. The adjustment of the blades shall be facilitated by an adjusting nut located at opposite ends of the blades. Blade ends shall be ground and rounded so that the adjusting nut, or washer when used, is not damaged by the blade ends. Length of blade shall be measured between adjusting nuts.

3.13.1.2 Adjusting nuts. Adjusting nuts shall have at least two flat sides, or other suitable means not requiring special tools or wrenches, for the expansion adjustment. The reamer body and the adjusting nuts shall be of carbon steel.

3.13.1.3 Threads. There shall be no evidence of burrs where the threads meet the blade slots. The shank diameter shall be of the largest practical diameter as determined by the root diameter of the adjusting thread (see 3.7).

3.13.2 Class 2, helical fluted (hand). The reamers shall have not less than three removable cutting segments with each segment having at least two helical cutting blades. The reamer shall have a straight, round shank with squared end. The segment mounting portion of the body shall be accurately machined or ground, concentric with the axis of the body, to insure uniform expansion of blades or segments throughout the range of adjustment. The reamer shall be constructed to permit an adjustment of not less than the minimum range, specified in table IV, above the nominal diameter listed. This minimum range shall be attainable with one set of blades or segments. The reamer assembly shall be free-cutting and capable of reaming round, straight, and smooth holes free of chatter marks within any size range. The reamers shall be similar to figure 3 and conform to table III.

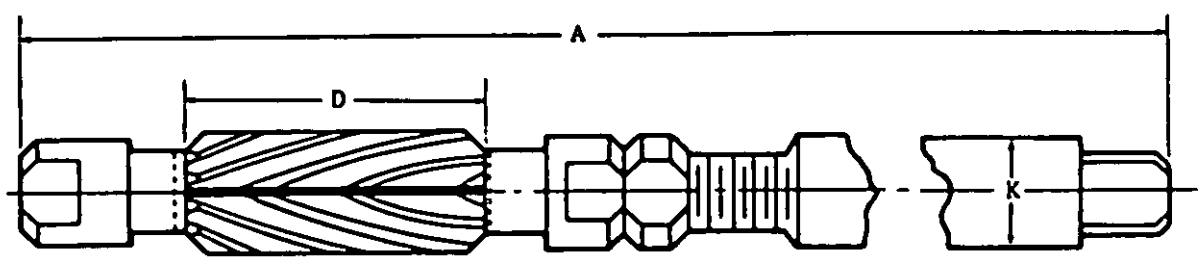


FIGURE 3. Type III, class 2, adjustable, inserted blade reamer, helical fluted (hand).

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TABLE III. Type III, class 2, adjustable, inserted blade reamer, helical fluted (hand).

Diameter	Range	Length of blades or segments (D) measured between adjusting nuts		Length, overall (A)		Shank diameter (K)
	Minimum	Minimum	Maximum	Minimum	Maximum	Maximum
Inches	Inch	Inches	Inches	Inches	Inches	Inches
1/2	0.035	1 3/8	2	5 1/2	8 3/4	0.480
17/32	.035	1 3/8	2	5 3/4	8 3/4	.480
9/16	.035	1 3/8	2	5 3/4	8 3/4	.480
19/32	.035	1 3/8	2 1/4	6 1/4	8 3/4	.480
5/8	.040	1 5/8	2 1/4	6 1/4	9 15/16	.610
21/32	.040	1 5/8	2 1/4	7	9 15/16	.610
11/16	.040	1 5/8	2 1/4	7	9 15/16	.610
23/32	.040	1 5/8	2 1/4	7	9 15/16	.610
3/4	.050	2 1/8	2 1/2	7 1/4	10 7/8	.720
25/32	.050	2 1/8	2 1/2	7 1/4	10 7/8	.720
13/16	.050	2 1/8	2 1/2	7 3/4	10 7/8	.720
27/32	.050	2 3/8	2 1/2	7 3/4	11 7/16	.840
7/8	.050	2 3/8	3	8 1/4	11 7/16	.840
29/32	.050	2 3/8	3	8 1/4	11 7/16	.840
15/16	.050	2 3/8	3	8 1/4	11 7/16	.840
31/32	.050	2 3/8	3	8 1/4	11 7/16	.840
1	.055	2 7/8	3	8 3/4	13 3/8	.970
1 1/32	.055	2 7/8	3	8 3/4	13 3/8	.970
1 1/16	.055	2 7/8	3	8 3/4	13 3/8	.970
1 3/32	.055	3 1/8	3 1/2	10	14	1.080
1 1/8	.055	3 1/8	3 1/2	10	14	1.080
1 5/32	.055	3 1/8	3 1/2	10	14	1.080
1 3/16	.055	3 1/8	3 1/2	11	14	1.080
1 7/32	.055	3 1/8	3 1/2	11	14	1.080
1 1/4	.055	3 3/8	3 1/2	11	14 5/8	1.230
1 9/32	.055	3 3/8	3 1/2	11	14 5/8	1.230
1 5/16	.055	3 3/8	3 1/2	11	14 5/8	1.230
1 11/32	.055	3 3/8	4	12	14 5/8	1.230
1 3/8	.055	3 3/8	4	12	14 5/8	1.230
1 13/32	.055	3 3/8	4	12	14 5/8	1.230
1 7/16	.055	3 3/8	4	12	14 5/8	1.230
1 15/32	.055	3 3/8	4	12	14 5/8	1.230
1 1/2	0.060	3 3/8	4	13	15 3/4	1.480

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3.13.2.1 Blades and segments. The blades and segments shall be accurately machined and ground in sets to fit the body of the reamer. The blades shall be right-hand cutting with left-hand helix or a combination of left-hand and right-hand helices as specified (see 6.2), and shall be uniform and of the same size and make of reamer to insure interchangeability. The cutting surfaces shall be concentric with the axis of the body in all positions throughout the range of adjustment and be capable of cutting on their entire length for the full range of adjustment. All blades and segments shall be chromium plated and properly marked in sets with corresponding minimum size. The blades and segments shall be of grade A steel.

3.13.2.2 Adjustment nuts, lock nuts (and floating collars, when provided). The outside diameter of adjusting nuts, lock nuts, and floating collars (when provided), shall be not less than 0.005 inch under minimum reamer size. Adjusting nuts and lock nuts shall have at least two flat sides, or other suitable means not requiring special tools or wrenches, for the expansion adjustment. The body of the reamer, adjusting nuts, lock nuts, and collars shall be of alloy steel.

3.13.3 Class 3, self-aligning (hand). The reamers shall be in accordance with the requirements for type III, class 2 reamers except that they shall also be furnished with a pilot and pilot bushing. The reamers shall be similar to figure 4 and conform to table IV.

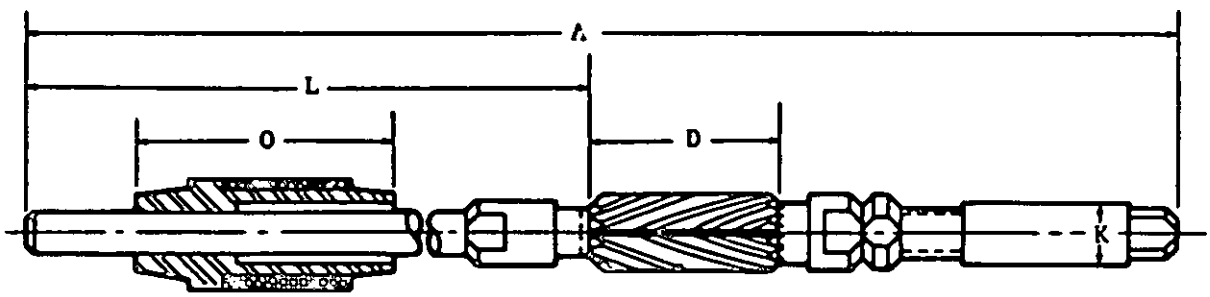


FIGURE 4. Type III, class 3, adjustable, inserted blade reamer, self-aligning (hand).

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TABLE IV. Type III, class 3, adjustable, inserted blade reamer, self-aligning (hand).

Diameter	Range	Length of blade or segments measured between adjusting nuts (D)		Length overall (A)		Length overall pilot (L)		Pilot bushing length		Shank Diameter (K)
	Minimum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Maximum
Inches	Inch	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/2	.035	1 3/8	2	9 1/4	14 7/8	4	6 13/16	2 3/4	2 7/8	.480
17/32	.035	1 3/8	2	9 1/4	14 7/8	4	6 13/16	2 3/4	2 7/8	.480
9/16	.035	1 3/8	2	9 1/4	14 7/8	4	6 13/16	2 3/4	2 7/8	.480
19/32	.035	1 3/8	2	9 1/4	14 7/8	4	6 13/16	2 3/4	2 7/8	.480
5/8	.040	1 5/8	2	9 1/4	17 1/16	4	7 15/16	2 3/4	2 7/8	.610
21/32	.040	1 5/8	2	9 1/4	17 1/16	4	7 15/16	2 3/4	2 7/8	.610
11/16	.040	1 5/8	2	12 1/4	17 1/16	4 1/2	7 15/16	2 3/4	2 7/8	.610
23/32	.040	1 5/8	2	12 1/4	17 1/16	4 1/2	7 15/16	2 3/4	2 7/8	.610
3/4	.050	2	2 1/8	12 1/4	18 1/4	4 1/2	8 7/16	2 7/8	3	.720
25/32	.050	2	2 1/8	12 1/4	18 1/4	4 1/2	8 7/16	2 7/8	3	.720
13/16	.050	2	2 1/8	12 1/4	18 1/4	4 1/2	8 7/16	2 7/8	3	.720
27/32	.050	2	2 3/8	12 1/4	19 3/4	4 1/2	9 7/16	2 7/8	3	.840
7/8	.050	2	2 3/8	12 1/4	19 3/4	4 1/2	9 7/16	2 7/8	3	.840
29/32	.050	2	2 3/8	12 1/4	19 3/4	4 1/2	9 7/16	2 7/8	3	.840
15/16	.050	2	2 3/8	12 1/4	19 3/4	4 1/2	9 7/16	2 7/8	3	.840
31/32	.050	2	2 3/8	12 1/4	19 3/4	4 1/2	9 7/16	2 7/8	3	.840
1	.055	2	2 7/8	12 1/4	21 3/4	4 1/2	9 9/16	2 7/8	3 1/4	.970
1 1/32	.055	2	2 7/8	12 1/4	21 3/4	6	9 9/16	2 7/8	3 1/4	.970
1 1/16	.055	2	2 7/8	12 1/4	21 3/4	6	9 9/16	2 7/8	3 1/4	.970
1 3/32	.055	2	3 1/8	12 1/4	22 9/16	6	9 13/16	2 7/8	3 5/8	1.080
1 1/8	.055	2	3 1/8	13	22 9/16	6 1/4	9 13/16	2 7/8	3 5/8	1.080
1 5/32	.055	3	3 1/8	13	22 9/16	6 1/4	9 13/16	2 7/8	3 5/8	1.080
1 3/16	.055	3	3 1/8	13	22 9/16	6 1/4	9 13/16	2 7/8	3 5/8	1.080
1 7/32	.055	3	3 1/8	13	22 9/16	6 1/4	9 13/16	2 7/8	3 5/8	1.080
1 1/4	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 9/32	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 5/16	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 11/32	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 3/8	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 13/32	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 7/16	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 15/32	.055	3	3 3/8	13	23 7/8	6 1/4	10 1/2	2 7/8	4 1/8	1.230
1 1/2	.060	3 5/16	3 7/16	25 7/8	26	11 1/2	11 5/8	4 1/4	4 3/8	1.480
1 5/8	.060	3 5/16	3 7/16	25 7/8	26	11 1/2	11 5/8	4 1/4	4 3/8	1.480

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3.13.3.1 Pilot and pilot bushing. A tapered pilot bushing shall be provided with each self-aligning hand reamer to facilitate self-alignment. The pilot hole and tapers on the bushing shall be concentric within 0.0005 inch tir. The hole shall be a sliding fit on the pilot with a 0.0003 to 0.0005 inch clearance. The length of taper pilot bushing shall be in accordance with the manufacturer's standard practice. The pilots shall be chromium plated.

3.13.4 Class 4, chucking, straight shank (machine). The reamers shall have not less than three removable cutting segments with each segment having at least two helical flutes, and shall be constructed to permit an adjustment of not less than that specified in table X with any one set of blades or segments. The reamer assembly shall be free-cutting and capable of reaming a round, straight, smooth hole free of chatter marks within any size range. The reamers shall be similar to figure 5 and shall conform to table V.

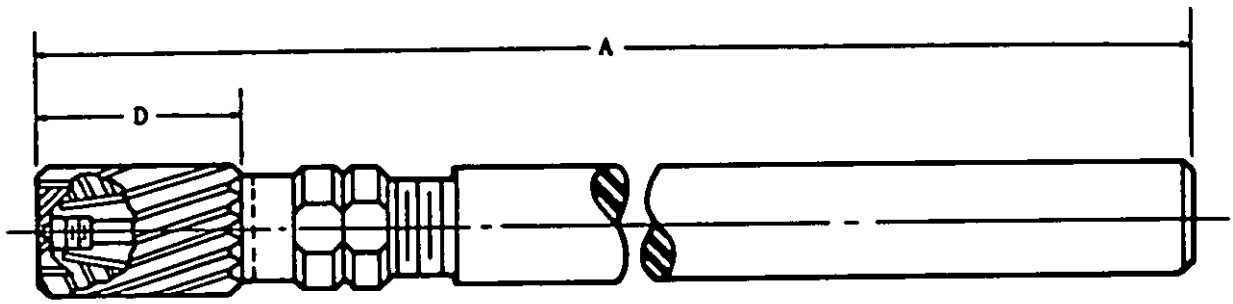


FIGURE 5. Type III, class 4, adjustable, inserted blade reamer, chucking, straight shank (machine).

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TABLE V. Type III, class 4, adjustable, inserted blade reamer, chucking, straight shank (machine).

Diameter	Range	Length of blade or segment (D)		Overall length (A)		Shank diameter	
	Minimum	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Inches	Inch	Inches	Inches	Inches	Inches	Inch	Inches
7/16	.009	7/8	1	7	8	3/8	3/8
15/32	.009	7/8	1	7	8	3/8	3/8
1/2	.009	15/16	1	7 1/2	8	7/16	7/16
17/32	.009	15/16	1	7 1/2	8	7/16	7/16
9/16	.009	15/16	1 1/8	7 1/2	8	7/16	7/16
19/32	.009	15/16	1 1/8	7 1/2	8	7/16	7/16
5/8	.012	1 3/16	1 1/4	7 1/2	9	9/16	9/16
21/32	.012	1 3/16	1 1/4	7 1/2	9	9/16	9/16
11/16	.012	1 3/16	1 1/4	7 1/2	9	9/16	9/16
23/32	.012	1 3/16	1 1/4	8 1/2	9	9/16	9/16
3/4	.027	1 5/16	1 3/8	8 1/2	9 1/2	5/8	5/8
25/32	.027	1 5/16	1 3/8	8 1/2	9 1/2	5/8	5/8
13/16	.027	1 5/16	1 3/8	8 1/2	9 1/2	5/8	5/8
27/32	.027	1 5/16	1 3/8	8 1/2	9 1/2	5/8	5/8
7/8	.033	1 3/8	1 1/2	8 1/2	10	3/4	3/4
29/32	.033	1 3/8	1 1/2	8 1/2	10	3/4	3/4
15/16	.033	1 3/8	1 1/2	8 1/2	10	11/16	3/4
31/32	.033	1 3/8	1 1/2	8 1/2	10	11/16	3/4
1	.033	1 9/16	1 5/8	8 1/2	10 1/2	11/16	7/8
1 1/32	.033	1 9/16	1 5/8	9 3/8	10 1/2	11/16	7/8
1 1/16	.033	1 9/16	1 5/8	9 3/8	10 1/2	11/16	7/8
1 3/32	.033	1 9/16	1 5/8	9 3/8	10 1/2	11/16	7/8
1 1/8	.033	1 9/16	1 3/4	10	10 1/2	11/16	7/8
1 5/32	.033	1 9/16	1 3/4	10	10 1/2	11/16	7/8
1 3/16	.033	1 9/16	1 3/4	10	11	11/16	7/8
1 7/32	.033	1 9/16	1 3/4	10	11	11/16	7/8
1 1/4	.034	1 1/2	2	10	12	3/4	1
1 9/32	.034	1 1/2	2	10	12	3/4	1
1 5/16	.034	1 1/2	2	10	12	3/4	1
1 11/32	.034	1 1/2	2	10 1/2	12 3/4	3/4	1
1 3/8	.034	1 1/2	2	10 1/2	12 3/4	3/4	1
1 13/32	.034	1 1/2	2	10 1/2	12 3/4	3/4	1
1 7/16	.034	1 1/2	2	10 1/2	12 3/4	7/8	1 1/4
1 15/32	.034	1 5/8	2	10 1/2	12 3/4	7/8	1 1/4

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3.13.4.1 Blades or segments. The blades or segments shall be accurately machined, hardened and ground in sets, and shall properly fit the segment-mounting portion of the body of the reamer. The blades or segments shall be right-hand cutting with left-hand helix, and shall be uniform for the same size and make of reamer to insure interchangeability. The cutting surfaces shall be concentric with the axis of the body in all positions throughout the range of adjustment and shall be capable of cutting on their entire length for the full range of expansion. They shall be properly marked in sets with corresponding minimum size and shall be chromium plated.

3.13.4.2 Adjusting nuts, lock nuts (and collar, when provided). The outside diameters of the adjusting nuts, lock nuts, and collar (when provided) shall be not less than 0.005 inch under the minimum reamer size. Adjusting nuts and lock nuts shall have at least two flat sides, or other suitable means not requiring special tools or wrenches, for the expansion adjustment.

3.13.4.3 Material. The body of the reamer, lock nuts, and adjusting mechanism shall be of alloy steel. The blades or segments shall be of grade A steel.

3.13.5 Class 5, chucking, taper shank (machine). The reamers shall conform to 3.13.4 except that class 5 reamers shall have tapered shanks, shall be similar to figure 6, and shall conform to table VI.

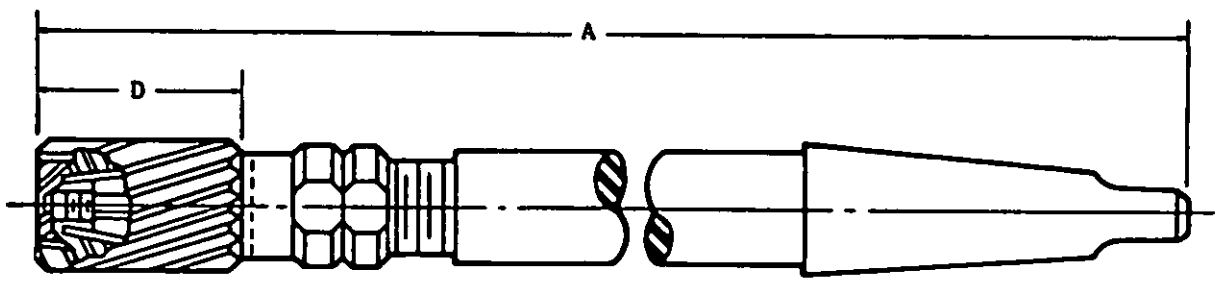


FIGURE 6. Type III, class 5, adjustable, inserted blade reamer, chucking, taper shank (machine).

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TABLE VI. Type III, class 5, adjustable, inserted blade reamer, chucking, taper shank (machine).

Diameter	Range	Length of blades or segment (D)		Overall length (A)		Number of Morse taper shank
	Minimum	Minimum	Maximum	Minimum	Maximum	
Inches	Inch	Inches	Inches	Inches	Inches	
7/16	.009	7/8	1	7	8	1
15/32	.009	7/8	1	7	8	1
1/2	.009	15/16	1	7 1/2	8	1
17/32	.009	15/16	1	7 1/2	8	1
9/16	.009	15/16	1 1/8	7 1/2	8	1
19/32	.009	15/16	1 1/8	7 1/2	8	1
5/8	.012	1 3/16	1 1/4	7 1/2	9	2
21/32	.012	1 3/16	1 1/4	7 1/2	9	2
11/16	.012	1 3/16	1 1/4	7 1/2	9	2
23/32	.012	1 3/16	1 1/4	8 1/2	9	2
3/4	.027	1 5/16	1 3/8	8 1/2	9 1/2	2
25/32	.027	1 5/16	1 3/8	8 1/2	9 1/2	2
13/16	.027	1 5/16	1 3/8	8 1/2	9 1/2	2
27/32	.027	1 5/16	1 3/8	8 1/2	9 1/2	2
7/8	.033	1 3/8	1 1/2	8 1/2	10	3
29/32	.033	1 3/8	1 1/2	8 1/2	10	3
15/16	.033	1 3/8	1 1/2	8 1/2	10	3
31/32	.033	1 3/8	1 1/2	8 1/2	10	3
1	.033	1 9/16	1 5/8	8 1/2	10 1/2	3
1 1/32	.033	1 9/16	1 5/8	9 3/8	10 1/2	3
1 1/16	.033	1 9/16	1 5/8	9 3/8	10 1/2	3
1 3/32	.033	1 9/16	1 5/8	9 3/8	10 1/2	3
1 1/8	.033	1 9/16	1 3/4	10	10 1/2	3
1 5/32	.033	1 9/16	1 3/4	10	10 1/2	3
1 3/16	.033	1 9/16	1 3/4	10	11	3
1 7/32	.033	1 9/16	1 3/4	10	11	3
1 1/4	.034	1 1/2	2	10	12	4
1 9/32	.034	1 1/2	2	10	12	4
1 5/16	.034	1 1/2	2	10	12	4
1 11/32	.034	1 1/2	2	10 1/2	12 3/4	4
1 3/8	.034	1 1/2	2	10 1/2	12 3/4	4
1 13/32	.034	1 1/2	2	10 1/2	12 3/4	4
1 7/16	.034	1 1/2	2	10 1/2	12 3/4	4
1 15/32	.034	1 5/8	2	10 1/2	12 3/4	4

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3.14 Type IV, pilots for type III, class 1, adjustable reamers. The pilot shall consist of a pilot body, pilot bushing, and retaining collar. The pilot hole and tapers of the bushing shall be concentric within 0.0005 inch tir. The bushing shall be double tapered and both ends shall take the full adjustment of the reamer. The bushing shall provide a sliding fit on the pilot body with a 0.0003 to 0.0005 inch clearance. Pilots shall be similar to figure 7 and conform to table VII.

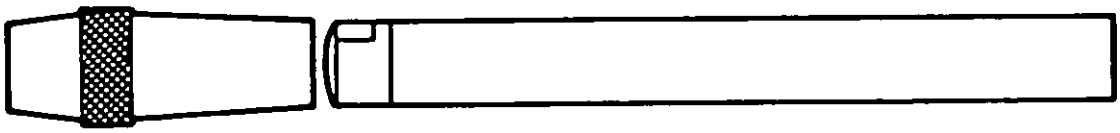


FIGURE 7. Type IV pilots for type III, class 1, adjustable reamers.

TABLE VII. Type IV, pilots for type III, class 1, adjustable reamers.

Size	Pilot diameter (nominal)	Overall length		Used with reamer size	Thread sizes
		Minimum	Maximum		
	Inches	Inches	Inches		
A	13/32	4 7/8	5 1/8	A	5/16-24
B	15/32	4 7/8	5 1/8	B	3/8-24
C	17/32	5	5 1/4	C	7/16-20
D	19/32	5 5/8	6	D	1/2-20
E	21/32	5 5/8	6	E	9/16-18
F	23/32	5 5/8	6	F	5/8-18
G	25/32	5 5/8	6	G	11/16-16
H	27/32	5 5/8	6	H	3/4-16
I	31/32	5 5/8	6	I	7/8-14
J	31/32	5 5/8	6	J	7/8-14
K	1 3/32	5 5/8	6	K	1-14

3.15 Type V, American National Standard taper socket (Morse). The cutting section of type V reamers shall be properly tapered for reaming American Standard sockets used for tools with American National Standard (Morse) taper shanks.

3.15.1 Class 1, straight shank (hand). The reamers shall have straight, round shanks with squared ends and shall be of grade A steel.

3.15.1.1 Style A, finishing. The reamers shall have straight flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.15.1.2 Style B, roughing. The reamers shall have straight flutes suitably notched for rough reaming. The reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2.

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3.15.2 Class 2, taper shank (machine). The reamers shall have American National Standard (Morse) taper shanks and shall be made of grade A steel. As specified (see 6.2), the reamers shall have straight or spiral flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.16 Type VI, taper, pipe. The reamers shall be of grade A steel, left-hand spiral flute, and shall have round shanks with squared ends. The cutting section of the reamers shall taper 3/4 inch per foot and shall be suitable for reaming holes to be tapped with American National taper pipe taps. Reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.17 Type VII, burring, pipe. The reamers shall be suitable for removing burrs from the inside edges of pipe, for countersinking, and for enlarging holes in sheet metal. The cutting edges shall be properly ground for the intended purpose. Reamers shall taper 30 degrees (included angle).

3.17.1 Class 1, ratchet type (hand).

3.17.1.1 Design. The reamers shall be of grade B steel and consist essentially of a lever, handle, ratchet mechanism, and reamer. The unit shall be furnished complete with an inserted tooth reamer or a solid cone reamer. All exposed parts except the reamer shall be painted, lacquered, enameled, or otherwise protected against corrosion. The reamers shall be similar to figure 8 and shall conform to table VIII.

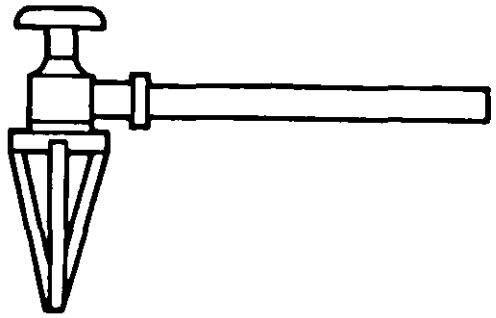


FIGURE 8. Type VII, class 1, burring, pipe reamer, ratchet type.

TABLE VIII. Type VII, class 1, burring, pipe reamer, ratchet type.

Capacity pipe	Distance from the centerline of reamer to end of lever (Minimum)	Weight (complete unit including reamer) (Maximum)
Inches	Inches	Pounds
1/8 to 2	10	7 3/4
3/8 to 3	10	9 1/2

3.17.1.2 Lever. The lever shall be made of cast iron or steel, be solid or hollow, and shall afford a comfortable grip for the operator. The lever shall be made integral with the ratchet collar or shall be provided with means for being securely fastened to the ratchet collar.

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3.17.1.3 Ratchet mechanism. The ratchet mechanism shall consist of a body with ratchet teeth, a pawl of steel hardened and held in contact with the teeth, and a ratchet collar. The teeth shall be enclosed within the ratchet collar. The reamer shall be designed to operate in a clockwise direction.

3.17.1.4 Handle. A suitable handle shall be provided, directly in line with the reamer, to enable the operator to support and guide the reamer during operation. The handle shall rotate about the ratchet body and reamer, and shall be attached in a manner to preclude its working loose under service conditions.

3.17.1.5 Reamer. The reamer shall consist of either a body fitted with not less than three blades, hardened and ground, or of a solid cone body with either straight or spiral flutes, as specified (see 6.2), hardened and tempered to a Rockwell hardness value of not less than C54 and not more than C58.

3.17.2 Class 3, bit brace shank (hand). Type VII, class 3 reamers shall be of grade B steel and have a bit brace shank. The reamers shall have a minimum of 8 to a maximum of 12 flutes, be similar to figure 9, and conform to table IX.

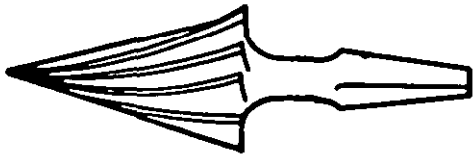


FIGURE 9. Type VII, class 3, style A, burring, pipe reamer, bit brace shank (hand) spiral fluted.

TABLE IX. Type VII, class 3, style A reamer.

Pipe capacity	Diameter (inches)	
	At point	At large end
Inches		
1/8 to 1	3/16	1 . 1/4
1/4 to 1 1/4	1/4	1 15/32
1/4 to 2	1/4	2 17/64

3.17.3 Class 4, straight shank (machine). The reamer shall be of grade A or grade B steel, have a minimum of 8 to a maximum of 12 spiral flutes, and a straight 1/2 inch diameter round shank. Straight shank reamers shall be similar to figure 10 and shall conform to table X.

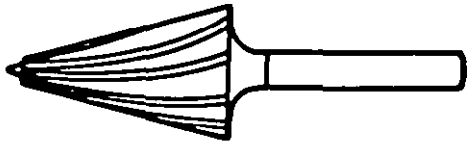


FIGURE 10. Type VII, class 4, burring, pipe reamer, straight shank (machine).

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TABLE X. Type VII, class 4 reamer.

Pipe capacity	Diameter (inches)	
	At point	At large end
Inches		
1/8 to 1	3/16	1 1/4
1/4 to 1 1/4	1/4	1 15/32
1/4 to 2	1/4	2 17/64

3.18 Type VIII, taper pin. The cutting section of type VIII reamers shall taper 1/4 inch per foot.

3.18.1 Class 1, straight fluted (hand). The reamers shall be of grade A steel, have straight flutes, straight round shanks with squared ends, and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.18.2 Class 2, spiral fluted (hand). The reamers shall be of grade A steel and shall have left-hand spiral flutes. Reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.18.3 Class 3, high spiral fluted (machine). The reamers shall be of grade A steel and have straight, round shanks and left-hand high spiral flutes cut at an angle from 30 degrees to 50 degrees with regard to the longitudinal axis of the reamer. Reamers shall be suitable for production reaming when used in a drill press or an electric or pneumatic drill. The reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.19 Type IX, bridge, helical fluted (machine). The reamers shall be of grade A steel and have left-hand helical flutes and American National Standard (Morse) taper shanks. The diameters of the cutting sections shall be ground the entire length. Reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.20 Type X, fluted, chucking (machine). The reamers shall have a slight chamfer at the entering end suitable for end cutting. A relief shall be ground for the full length of each land.

3.20.1 Class 1, taper shank. The reamers shall be of grade A steel and have American National Standard (Morse) taper shanks. The body of the reamer between the flutes and shank shall be designed to permit the reamer to enter the hole up to the large end of the shank.

3.20.1.1 Style A, straight fluted. The reamers shall have straight flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.20.1.2 Style B, helical fluted. The reamers shall have helical flutes, right or left-hand helix as specified (see 6.2), right-hand cut, and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.20.2 Class 2, straight shank. The reamers shall be of grade A steel and have straight round shanks.

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3.20.2.1 Style A, straight fluted. The reamers shall have straight flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.20.2.2 Style B, helical fluted. The reamers shall have helical flutes, right or left-hand helix as specified (see 6.2), right-hand cut, and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.21 Type XII jobbers' (machine). The reamers shall be of grade A steel and have straight flutes and American National Standard (Morse) taper shanks. Reamers shall have a slight chamfer for end cutting. Flutes shall be designed for machine use. Reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.22 Type XIII, repairman's T-handle (hand). The reamers shall be of grade B steel and shall have straight flutes and a solid T-handle. Reamers shall have a 1/8 inch maximum diameter at the small end with a pointed tip. The large diameter shall be 1/2 inch. The cutting section of these reamers shall taper 1.250 inches per foot and shall be suitable for enlarging holes in thin metal. The reamers shall be similar to figure 11.

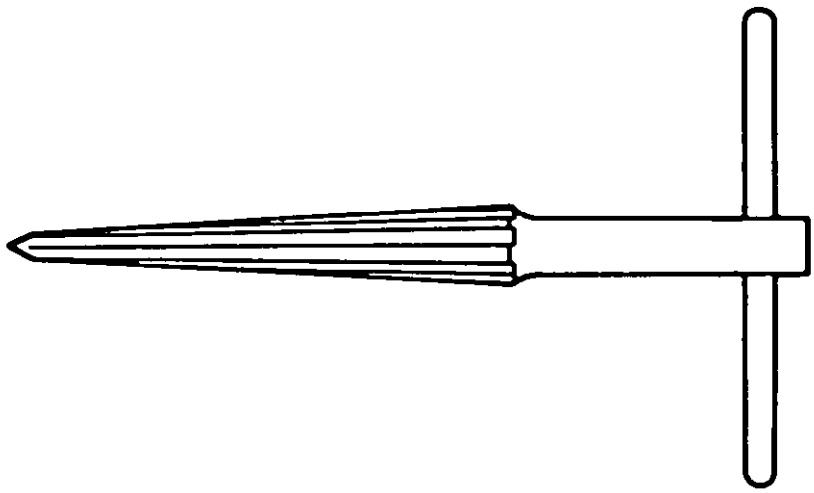


FIGURE 11. Type XIII, repairman's T-handle (hand) reamer.

3.23 Type XIV, classes 1 and 2, Brown and Sharpe taper socket (hand). The reamers shall have a straight or spiral fluted taper section as specified (see 6.2), and straight, round shanks with squared ends. The cutting section of these reamers shall be properly tapered for reaming Brown and Sharpe standard sockets used for tools with Brown and Sharpe taper shanks. Reamers shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.24 Type XV, shell (machine). The reamers shall be of grade A steel designed for use with arbors. The arbor holes shall taper 0.125 inch per foot and shall properly fit the arbors for shell reamers, type XVI (see 3.25). The corners at the bottom of the driving slots shall be rounded.

3.24.1 Class 1, straight fluted. Type XV, class 1, reamers shall have straight flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.24.2 Class 2, left-hand helical fluted. The reamers shall have left-hand helical flutes and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

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3.25 Type XVI, arbors (for type XV reamers). The end of the arbor fitting the reamer shall properly fit arbor holes in type XV reamers as specified in 3.24, tapering 0.125 inch per foot. The arbors shall be furnished with a driving collar which shall be pressed or shrunk in place and pinned to the arbor by means of a taper pin or other standard manufacturing practice. The lugs of the collar shall be centrally located within 0.002 inch of the center line of the arbor and shall be designed to engage driving slots of the type XV reamers. Arbors and collars shall be of grade A or B steel. Carbon steel arbors shall be hardened and ground (see 3.6). Arbors of grade A steel shall be hardened and ground in accordance with the manufacturer's standard practice.

3.25.1 Class 1, taper shank. The arbors shall have American National Standard (Morse) taper shanks and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.25.2 Class 2, straight shank. The arbors shall have straight shanks and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.26 Type XVIII, car reamer, left-hand, helical fluted, taper shank (machine). The reamers shall be of grade A steel, have American National Standard (Morse) taper shanks, and shall conform to ANSI B94.2 for the dimensions specified (see 6.2).

3.27 Reamer sets. Sets of reamers of the types, classes, grades, and styles covered by this specification shall conform to the applicable requirements of this specification for the respective reamers included. Each reamer in a set shall be made by the same manufacturer. Each set shall be packed in a wood, plastic, or metal case which shall have a compartment for each reamer to make for convenient selection, to prevent damage to cutting edges by contact, and to provide a compact method of storage when not in use. The wood and metal cases shall be fitted with at least two hinges, or a continuous hinge, and at least one shutter hook or similar hardware to secure the case in a closed position. All hardware shall be corrosion resistant or have a corrosion resistant coating. If the case is made of metal, it shall be corrosion resistant or have a corrosion resistant coating or finish. If the case is of wood, it shall have an oil finished interior. The exterior shall be varnished or shellacked. If the case is made of plastic, it shall be a double layer, impact resistant polymer with a continuous hinge and one or more snap-type latches. All cases shall be marked on the cover with the appropriate reamer sizes of the respective size ranges of each reamer provided in the set.

3.28 Workmanship. Standards of workmanship shall assure that the reamers, arbors, and pilots shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection and tests. When specified (see 6.2), the first article shall be inspected for defects in table XI and subjected to all the applicable tests in 4.4. One or more defects or failure to meet all the applicable test requirements of 4.4 shall be cause for rejection.

TABLE XI. Test holes for type III, classes 1, 2, and 3.

Class	Nominal size reamer
	Size letters
1	8/A through 2/A A through G H through O
	Inches
2 and 3	1/2 through 25/32 13/16 through 1 1/8 1 5/32 through 1 5/8

4.2.2 Quality conformance inspection. Each sample taken from the inspection lot shall be examined for compliance with 4.5 and subjected to the applicable reaming tests in 4.4. Sampling shall be in accordance with FED-STD-358 for all examinations and tests except the reaming tests. Sampling for reaming shall be in accordance with MIL-STD-105, based on normal inspection at Inspection Level S-2 with an Acceptable Quality (AQL) of 4.0 percent defective.

4.3 Inspection lot. All reamers or reamer of the same type, class, style, grade, and size which have been packaged, packed, marked for shipment, and offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.4 Test procedure.

4.4.1 Hardness test and defective material inspection. Each sample reamer, arbor, or pilot, as applicable, shall be tested for varification of hardness requirement specified in 3.6. Magnetic particle inspection, X-ray, or other approved means shall be employed to inspect the reamers, arbors, or pilots for hidden breaks and quench cracks.

4.4.2 Performance tests. The material for testing the reamers shall be steel conforming to FED-STD-66, AISI Number 4140 or 4150, with a Brinell hardness value of not less than 235 nor more than 285. Test material thickness shall be as specified for each type and class reamer. Unless otherwise specified herein, machine reamers shall be mounted in a floating tool holder and performance tested in a drill press or other suitable machine using flood coolant for the reaming

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tests. Except as otherwise specified, the reaming speed should be approximately 85 surface feet per minute (sfm) and the feed per revolution directly proportionate to the reamer diameter i.e., 0.005 inch feed for 0.250 diameter reamers and 0.012 inch feed for 0.750 diameter reamers. All finish-reamed holes, both machine and hand reamers, shall be within the tolerance limits specified for applicable reamer.

4.4.2.1 Type I (hand reamer) test. The reamers shall be tested by hand reaming ten holes under standard machine shop practice for hand reamers. The reamers shall be tested in material that is three times thicker than the diameter of the reamer or 1-1/2 inches, whichever is less, except for Class 3 reamers which shall be tested in 1/2 inch thick material. The test holes shall be drilled or bored to a diameter which shall leave stock remaining approximately three percent of the reamer diameter for reamers up to 3/4 inch and two percent for reamers 3/4 inch and larger. (For example, a 21/32 inch reamer would be tested by drilling a 41/64 inch hole and reaming with the 21/32 reamer.) The reamed hole shall be inspected for finish and parallelism. After reaming the third hole, the reamed diameter shall be recorded. The diameter of the tenth hole shall also be recorded. The recorded diameters shall not vary more than plus zero and minus 0.0002 inch. For sets, three representative sizes shall be subjected to the above tests.

4.4.2.2 Type I, class 3 (hand reamer) test. The reamer shall be tested by hand reaming a minimum of ten acceptable holes in 1/2 inch thick steel. The test holes shall be prepared 0.002 to 0.003 inch smaller than the size hole desired. A straight plug gage shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.

4.4.2.3 Type II, classes 1 and 2 (hand reamer) test. The reamer shall be tested by hand reaming a minimum of ten acceptable holes in 1/2 inch thick steel for reamers 3/4 inch diameter and smaller and in one inch thick steel for reamers over 3/4 inch diameter. The test holes shall be prepared 0.002 to 0.003 inch smaller than the size of the hole desired. The reamer shall make a series of cuts, not exceeding 0.001 inch, to remove the desired amount of metal.

4.4.2.4 Type III, classes 1, 2 and 3 (hand reamer) test. The reamers shall be tested by reaming ten holes of each size specified in 1/2 inch thick steel. The test holes shall be prepared by drilling or boring which shall leave remaining stock approximately three percent of the diameter within the range of the reamers up to 3/4 inch in diameter and approximately two percent for reamers 3/4 inch and larger. (For example, a letter size C reamer would be tested by drilling a 41/64 inch hole and reaming to 21/32 inch.) The reamed hole shall be inspected for surface finish and parallelism. After reaming the third hole, the reamed diameter shall be recorded. The diameter of the tenth hole shall also be recorded. The diameters shall not vary more than plus zero and minus 0.0002 inch. For sets, one reamer in each of the size ranges specified in table XI, for the applicable class of reamer, shall be tested.

4.4.2.5 Type III, classes 4 and 5 (machine reamer) test. The machine reamers shall be tested by reaming a minimum of 100 acceptable holes in 1-1/4 inch thick steel. The test holes shall be prepared 0.012 to 0.015 inch smaller than the size of the hole desired. The reamer shall be operated at a speed in revolutions per minute that gives approximately 80 sfm for any size tested with a feed rate of 1-1/2 percent of the reamer diameter per revolution. Reaming shall

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produce smooth and round holes. A straight plug gage the size of the finished hole shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.

4.4.2.6 Type V, class 1 taper reamer (hand) test. The reamer shall be tested by hand reaming a minimum of three acceptable holes. The test shall be conducted in a steel plate of at least one inch thickness in which test holes have been prepared with a taper approximating that of the reamer taper. Reaming shall produce a smooth, accurately tapered hole.

4.4.2.7 Type V, class 2 taper reamer (machine) test. The machine reamer shall be tested by reaming a minimum of five acceptable holes in 1-1/4 inch thick steel. The starting hole shall be 0.002 inch to 0.003 inch smaller than the small diameter of the finish-reamed hole. Reaming shall produce a smooth, accurately tapered hole.

4.4.2.8 Type VI, taper, pipe reamer test. The reamer shall be tested by reaming a minimum of three acceptable holes. The thickness of the steel plate shall be at least 1/2 inch. Reaming shall produce a smooth, accurately tapered hole.

4.4.2.9 Type VII, burring pipe reamer test. The reamer shall be subjected to test, as normally operated, by reaming at least two samples each of pipe within the capacity of the size reamer specified. The finish of the countersink in the sample pipes shall be smooth and free of burrs.

4.4.2.10 Type VIII, classes 1 and 2 (hand reamer) test. The reamer shall be tested by hand reaming a minimum of 25 acceptable holes using normal machine shop practice for hand reamers. The test material shall be at least 1/2 inch thick. The starting hole shall be 0.002 to 0.003 inch smaller than the small diameter of the finish-reamed hole. Failure of the reamer shall be considered to have occurred if the reamer fails to cut at any time before the minimum number of acceptable holes has been attained. For sets, three representative sizes shall be subjected to the above tests.

4.4.2.11 Type VIII, class 3 (machine) reamer test. The machine reamer shall be tested by reaming a minimum of 100 acceptable holes. The starting hole shall be 0.002 to 0.003 inch smaller than the small diameter of the finish-reamed hole. The reamer shall be operated at a speed equivalent to 60 sfm for the large diameter, with a feed in inches per revolution equal to one percent of the large diameter. The test material shall be at least 1/2 inch thick. Reaming shall produce a smooth, round, accurately tapered hole. Failure of the reamer shall be considered to have occurred if the reamer fails to cut at any time before the minimum number of acceptable holes has been obtained.

4.4.2.12 Type IX, bridge reamer (machine) test. The reamer shall be tested by reaming (without lubricant or coolant) a minimum of 50 acceptable holes. The test holes shall be prepared 0.062 to 0.065 inch smaller than the size of hole desired. The test material shall be at least 1/2 inch thick. The reamer shall be operated in a portable pneumatic drill at not less than 270 nor more than 280 revolutions per minute during the cutting operation. The total feed pressure applied to the reamer shall not exceed 37-1/2 pounds dead weight. Acceptable holes shall be holes which are smooth, free from burrs, and free from discoloration due to heat.

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4.4.2.13 Type X, classes 1 and 2 (machine reamer) test. The reamers shall be tested by reaming a minimum of 150 acceptable holes in 1-1/4 inch thick steel for reamers 7/16 inch diameter and larger, and in 1/2 inch thick steel for reamers having less than 7/16 inch diameter. The test holes shall be prepared by leaving reaming stock of three percent for reamers up to 3/4 inch diameter, and two percent for reamers 3/4 inch and larger. The reamer shall be operated at a surface speed equivalent to approximately 65 sfm with a feed rate in inches per revolution equivalent to 1.5 percent of reamer diameter. Reaming shall produce smooth and round holes. A straight plug gage the size of the finished hole shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.

4.4.2.14 Type XII, jobbers' (machine) reamer test. The reamer shall be tested by reaming a minimum of 100 acceptable holes in 1-1/4 inch thick steel. The test holes shall be prepared by leaving reaming stock equivalent to three percent of the reamer diameter. The reamer shall be operated at a surface speed equivalent to approximately 65 sfm with a feed rate in inches per revolution equivalent to 1.5 percent of reamer diameter. Reaming shall produce smooth and round holes. A straight plug gage the size of the finished hole shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.

4.4.2.15 Type XIII, repairman's T-handle reamer (hand) test. The reamer shall be tested by hand reaming a minimum of three acceptable holes in steel at least 1/8 inch thick. The test shall consist of enlarging the initial hole a minimum of 50 percent of original diameter without regard to taper produced.

4.4.2.16 Type XIV, Brown and Sharpe taper socket reamer (hand) test. The test shall be conducted in a steel plate at least one inch thick in which test holes have been prepared with a taper approximately that of the reamer taper. Reaming shall produce a minimum of three smooth and accurately tapered holes.

4.4.2.17 Type XV, shell reamer (machine) test. The reamer shall be tested by reaming a minimum of 50 acceptable holes in 1-1/4 inch thick steel, using type XVI arbors. The test holes shall be prepared by leaving stock, for reaming, equivalent to three percent of the reamer diameter. The reamer shall be operated at revolutions equivalent to 65 sfm with a feed rate in inches per revolution equivalent to 1.5 percent of reamer diameter. Reaming shall produce smooth and round holes. A straight plug gage the size of the finished hole shall be used to measure the size of the hole. Failure of the reamer shall be considered to have occurred when the plug gage will not enter the hole before the minimum number of acceptable holes is reached.

4.4.2.18 Type XVIII, car reamer (machine) test. The reamer shall be tested to meet the requirements of the type IX bridge reamer test.

4.5 Inspection of preparation for delivery. Preservation, packing, and marking for shipment and storage shall be inspected to determine compliance with Section 5.

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5 PREPARATION FOR DELIVERY

5.1 Preservation, packing, and marking. When specified (see 6.2), Level A preservation and Levels A or B packing and marking shall be in accordance with PPP-P-40.

5.2 Industrial preservation, packing, and marking. When specified (see 6.2), preservation, packing, and marking shall conform to the requirements of ASTM D3951-82.

5.3 Civil agency marking. When specified (see 6.2), marking for shipment for civil agencies shall be in accordance with FED-STD-123.

6 NOTES

6.1 Intended use. Reamers covered in this specification are intended for burring, reaming, and aligning operations on various hole diameters, tapered holes, and hole lengths to specified tolerances.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style of reamers required (see 1.2.1).
(c) First article, when required (see 3.1).
(d) Dimensions required for reamers, pilots, or arbors (see applicable tables).
(e) Pilots for type III, class 1, reamers when required (see 3.13.1).
(f) Right-hand or combination right and left-hand helical fluted type III, classes 2 and 3 reamers, when required (see 3.13.2.1 and 3.13.3).
(g) Straight or spiral fluted type V, class 2 reamers, as required (see 3.15.2) and type VII, class 1, as required (see 3.17.1.5).
(h) Right or left-hand helix type X, class 1 and 2, style B reamers, when required (see 3.20.1.2 and 3.20.2.2).
(i) Straight or spiral flutes for type XIV, as required (see 3.23).
(j) First article inspection and tests, when required (see 4.2.1).
(k) Selection of applicable levels of preservation, packing, and marking (see 5.1 and 5.2).
(l) Civil agency shipment marking when required (see 5.3).

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MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review activities

Army - GL
Air Force - 84

User activity

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:
Army - AL

DoD Project 5110-0296

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GGG-R-180E	Reamers, Hand and Machine: Arbors, Shell Reamer; and Pilots, Adjustable Hand Reamer

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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